

DAWOOD UNIVERSITY OF ENGINEERING AND TECHNOLOGY

23F BSAI

3rd Year, 6th Semester

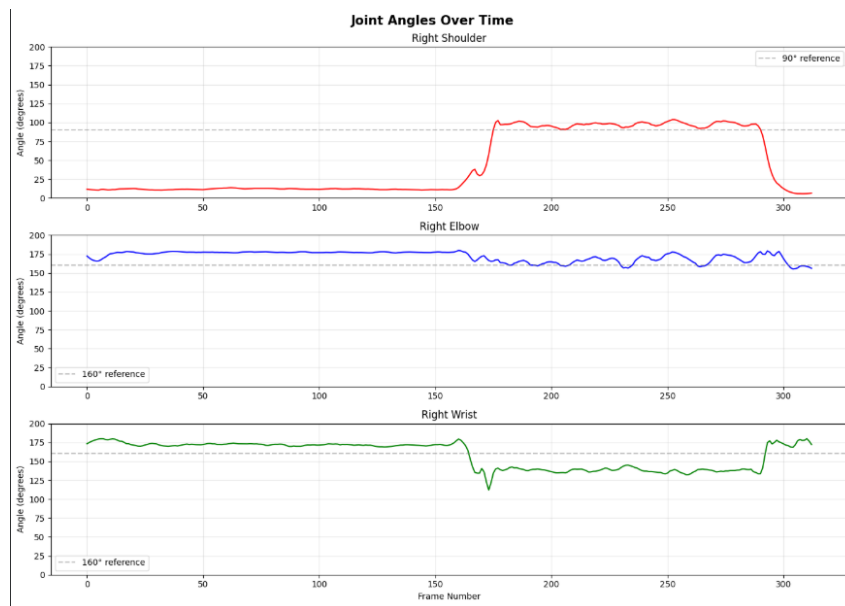
NAME:	SABIHA PIRZADAH
ROLL NO.:	23-AI-84
COURSE:	COMPUTER VISION
TITLE:	CCP ASSIGNMENT REPORT
TOPIC:	POSE DETECTION USING MediaPipe
DATE:	23/05/2026

Assignment Report

Angle Computation Report:

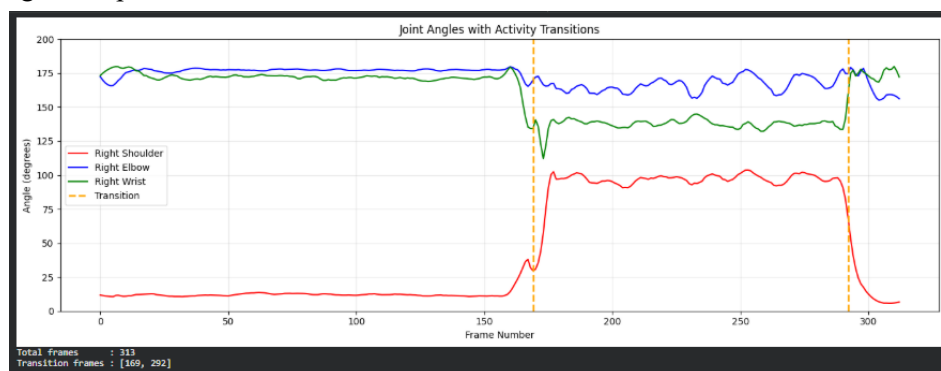
MediaPipe Pose was used to extract 33 body keypoints from each frame of the video. Since raw keypoint coordinates contain frame-to-frame jitter caused by model uncertainty, a Savitzky-Golay filter (window=11, polyorder=2) was applied to smooth the coordinates before any angle computation. This filter fits a polynomial curve through a sliding window of frames, preserving the shape of real motion while removing noise.

Angles were computed at three joints on the right arm, the shoulder, elbow, and wrist. For each joint, two vectors were formed from the joint outward toward its neighboring keypoints. The angle between these vectors was then calculated using the dot product formula and converted from radians to degrees



Rule-Based Classification Logic

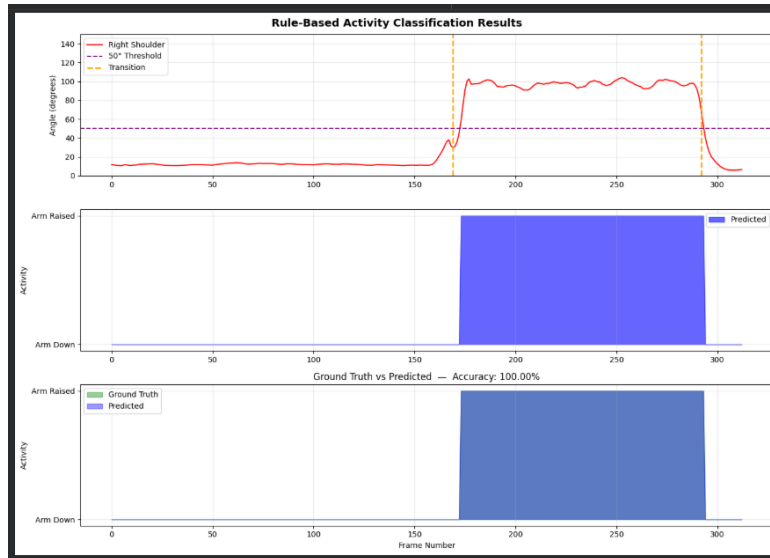
After plotting all three joint angles over time, the right shoulder angle showed the most meaningful variation across the video, ranging from 5° to 103° . The elbow and wrist remained mostly straight throughout, averaging 172° and 158° respectively, making them unsuitable as primary classifiers. A simple threshold rule was designed based on the shoulder angle. If the shoulder angle was greater than or equal to 50° , the frame was classified as "Arm Raised". Otherwise it was classified as "Arm Down". This threshold was chosen because it sits cleanly between the two observed angle ranges and produced no borderline cases in this video



Accuracy Results

The classifier was run across all 313 frames of the video. 192 frames were classified as Arm Down and 121 frames as Arm Raised. Transition frames were identified at frame 169 (arm going up) and frame 292 (arm coming down).

Manual ground truth labels were created based on the same transition frames observed in Task 2. The classifier achieved 100% accuracy against these labels. It should be noted that since the ground truth was derived from the same transitions the classifier detected, the 100% result reflects perfect internal consistency rather than an independent evaluation



Observation:

- ♣ The shoulder angle proved to be the most informative feature the elbow and wrist remained nearly straight throughout, confirming that the activity was driven entirely by arm elevation. This suggests that for any arm-raising activity, shoulder angle alone is sufficient for reliable classification.
- ♣ Smoothing was a necessary preprocessing step rather than an optional one. Raw keypoint coordinates contain inherent noise from the neural network's frame-by-frame uncertainty, and computing angles directly from noisy coordinates would propagate that error into the classifier. Clean inputs produced a clean decision boundary.
- ♣ The 50° threshold worked because the two activities occupied clearly separated angle ranges with no overlap. In more complex scenarios where activities share similar joint configurations, a rule-based approach would need to be replaced by a learned classifier that considers multiple joints and their relationships simultaneously.

